

Bit Error Rate Calculation for OFDM Direct Conversion Receivers with Synchronization and Channel Estimation Imperfections

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Abstract—In this paper we present an analytical approach to evaluate the M-QAM bit error rate (BER) of OFDM direct conversion receivers subject to carrier frequency offset (CFO), channel estimation error, outdated channel state information and flat receiver I/Q imbalance in Rayleigh time and frequency selective fading channels. Based on correct modeling of the correlation between channel estimates and received signals with carrier frequency offset and receiver I/Q imbalance, the bit error rate can be numerically evaluated by averaging bit error rates on different subcarriers using an analytical expression of triple integrals. The results illustrate that the analysis can approximate the simulative performance very accurately if the power delay profile of fading channels, the receiver I/Q imbalance parameters and carrier frequency offset are known.

I. INTRODUCTION

Orthogonal Frequency Division Multiplexing (OFDM) is a widely applied technique for wireless communications, which enables simple one-tap equalization by cyclic prefix insertion. Conversely, the sensitivity of OFDM systems to carrier frequency offset (CFO) is higher than that of single-carrier systems. In present OFDM standards, such as IEEE802.11a/g or DVB-T, preamble (or pilots) are used to estimate and compensate the carrier frequency offset (CFO) and channel impulse response. Unfortunately, after CFO estimation and compensation, the residual carrier frequency offset still destroys the orthogonality of the received OFDM signals and corrupts channel estimates, which worsen further the bit error rate of OFDM systems during the equalization process. In the literature, the effects of carrier frequency offset on symbol/bit error rate are mostly investigated under the assumption of perfect channel knowledge [4] [3] [5]. Additionally, the authors of [1] tried to analyze the joint effects of carrier frequency offset and channel estimation error on uncoded bit error rate for OFDM systems, but the assumption in [1] (channel estimates are uncorrelated to channel estimation error or uncorrelated to received signals) does not hold in real OFDM systems, especially when carrier frequency offset is large. Additionally, receiver I/Q imbalance has been identified as one of the most serious concerns in the practical implementation of direct conversion receiver architectures. Since,

transmitter I/Q imbalance is known to have much smaller effects on OFDM systems [6] than receiver I/Q imbalance, this impairment is not considered in this paper. The authors of [9] investigated the effect of receiver I/Q imbalance on OFDM systems for frequency selective fading channels under the assumption of perfect channel knowledge and perfect receiver synchronization. Additionally, in order to cope with this impairment, the authors of [8] proposed a digital I/Q imbalance compensation method. To our best knowledge, there is currently no work on the analytical BER calculation for OFDM systems under the aggregate effect of all the mentioned impairments. Finally we have to state that the knowledge about the quantitative relationship between receiver parameters (such as I/Q imbalance, channel estimation error and residual CFO) and performance metrics such as bit error rate is essential for the design and dimensioning of OFDM systems and receiver algorithms.

In this paper, after introducing the OFDM system model (Sec.II) and the probability density function analysis [7] (Sec.III), it will be explained how to model the correlation between channel estimates and received/impaired signals for deriving the bit error rates of OFDM systems with carrier frequency offset and I/Q imbalance in Rayleigh frequency and time selective fading channels (Sec.IV). It should be noted, that the terms bit error rate and bit error probability are used with equal meaning. This is due to the fact, that the bit error rate converges toward bit error probability with increasing observation time in an stationary environment. Finally we end up with the verification of the analytical results in Sec.V and Conclusions in Sec.VI.

II. OFDM SYSTEM MODEL

We consider an OFDM system with N-point FFT. The data is M-QAM modulated to different OFDM data subcarriers, then transformed to a time domain signal by IFFT operation and prepended by a cyclic prefix, which is chosen to be longer than the maximal channel impulse response (CIR) length L . The sampled discrete complex baseband signal for the l th subcarrier after the receiver FFT processing can be written

as

$$Y_l = X_l H_l + W_l \quad (1)$$

where X_l represents the transmitted complex QAM modulated symbol on subcarrier l and W_l is a complex Gaussian noise sample. The coefficient H_l denotes the frequency domain channel transfer function on subcarrier l , which is the discrete Fourier transform of the CIR $h(\tau)$ with maximal L taps

$$H_l = \sum_{\tau=0}^{L-1} h(\tau) e^{-j2\pi l\tau/N} \quad (2)$$

In this paper, it is assumed that the residual carrier frequency offset (after frequency synchronization) is a given deterministic value. A one-OFDM-symbol-long preamble is used for channel estimation. Furthermore, static channel characteristics during one OFDM symbol are assumed. The sampled CFO-impaired discrete complex baseband signal for the l th subcarrier after the receiver FFT processing can be written as

$$Y_l = X_l H_l I(0) + \sum_{k=-\frac{N}{2}-1}^{\frac{N}{2}-1} X_k H_k I(k-l) + W_l \quad (3)$$

where the coefficient $I(k-l)$ represents the impact of the received signal at subcarrier k on the received signal at subcarrier l due to the residual carrier frequency offset [3]

$$I(k-l) = e^{j\pi((k-l)+\Delta f)(1-1/N)} \frac{\sin(\pi((k-l)+\Delta f))}{N \sin(\pi((k-l)+\Delta f)/N)}$$

where Δf is the residual carrier frequency offset normalized to the subcarrier spacing. In addition, later in this paper the summation $\sum_{k=-\frac{N}{2}-1}^{\frac{N}{2}-1}$ will be abbreviated as $\sum_{k \neq l}$. In Eq.(3) we can see, that residual CFO causes a phase rotation of the received signal ($I(0)$) and inter carrier interference (ICI). Furthermore there is a time variant common phase shift for all subcarriers due to Δf as given in [5] that is not modeled here. This is due to the fact, that this time variant common phase term is considered to be robustly estimated and compensated by continuous pilots that are inserted in the OFDM data symbols.

I/Q imbalance of direct conversion OFDM receivers directly translates to a mutual interference between each pair of subcarriers located symmetrically with respect to the DC carrier [8]. Hence, the received signal Y_l at subcarrier l is interfered by the received signal Y_{-l} at subcarrier $-l$, and vice versa. Therefore, the undesirable leakage due to I/Q imbalance can be modeled by [8], [10]

$$\tilde{Y}_l = Y_l + K_l Y_{-l}^* \quad (4)$$

where $(\cdot)^*$ represents the complex conjugation and K_l denotes a complex-valued weighting factor that is determined by the receiver phase and gain imbalance [8]. The image rejection capabilities of the receiver on subcarrier l can be expressed in terms of *image rejection ratio* - *IRR* given by

$$\text{IRR}_l = \left| \frac{1}{K_l} \right|^2 \quad (5)$$

In this paper we consider *flat* I/Q imbalance which simply means $\text{IRR}_l = \text{IRR}$, $\forall l$. Subsequently, we consider preamble-based Least Square (LS) frequency domain channel estimation to obtain the channel state information on subcarrier l .

$$\hat{H}_l = \frac{\tilde{Y}_{P,l}}{X_{P,l}} = \frac{I(0)H_l + \sum_{k \neq l} X_{P,k} H_k I(k-l) + W'_l}{X_{P,l}} + K_l \frac{\sum_m X_{P,m}^* H_m^* I^*(m+l) + W'_{-l}}{X_{P,l}} \quad (6)$$

followed by the frequency domain zero-forcing equalization before data detection

$$Z_l = \frac{\tilde{Y}_l}{\hat{H}_l} \quad (7)$$

where $X_{P,l}$ and $\tilde{Y}_{P,l}$ denote the transmitted and received preamble symbol on subcarrier l . The Gaussian noise of preamble part W'_l has the same variance as W_l of the data part ($\sigma_{W'_l}^2 = \sigma_{W_l}^2$). The power of preamble signals and the average power of transmitted data signals on all subcarriers is normalized to one ($|X_P|^2 = \sigma_X^2 = 1$). The derived analytical expressions for bit error and symbol error rates are compared with appropriate simulation results of Sec.V. In these numerical examples, we consider an IEEE 802.11a like OFDM system [2] with 64-point FFT. The data is 4-QAM-modulated to different subcarriers, then transformed to the time domain signal by IFFT operation and prepended by a 16-tap long cyclic prefix. The data is randomly generated and one OFDM pilot symbol was used for channel estimation. The used BPSK pilot data in the frequency domain is given by

$$X_{P,l} = (-1)^l \text{ for subcarrier index } l = [-26 : 1 : 26] \quad l \neq 0$$

A. Mobile Channel Characteristics

Although our analysis is not limited to a specific type of frequency selective channel, frequency selective fading channels having an exponential power delay profile (PDP) are considered in this paper with

$$\sigma_\tau^2 = \frac{1}{C} e^{-D\tau/L} \quad , \quad \tau = 0, 1, \dots, L-1$$

where $\sigma_\tau^2 = E\{|h_\tau|^2\}$ and the factor $C = \sum_{\tau=0}^{L-1} e^{-D\tau/L}$ is chosen to normalize the PDP as $\sum_{\tau=0}^{L-1} \sigma_\tau^2 = 1$ (or $\sigma_H^2 = 1$ on all subcarriers). The channel taps h_τ are assumed to be complex Gaussian RV with uncorrelated real and imaginary parts. Additionally, the maximal channel impulse response is kept shorter/equal than the cyclic prefix ($L \leq 16$). The cross-correlation coefficient of channel transfer functions between subcarrier k and l in frequency selective fading channels are defined

$$r_{k,l} = \frac{E\{H_k H_l^*\}}{\sigma_H^2} \neq 1 \quad \forall k \neq l \quad (8)$$

where the channel gains σ_H^2 are the same for all subcarriers. Assuming mutual uncorrelated channel taps of the CIR and applying Eq. (2), one get

$$r_{k,l} = \sum_{\tau=0}^{L-1} \sigma_\tau^2 e^{-j2\pi(k-l)\tau/N} \quad (9)$$

The cross correlation properties of the channel coefficients is written as follows

$$H_k = r_{k,l}H_l + V_{k,l} \quad (10)$$

where V_k a complex zero-mean Gaussian with variance $\sigma_{V_{k,l}}^2 = \sigma_H^2(1 - |r_{k,l}|^2)$ and $E\{V_{k,l}H_l^*\} = 0$.

In current OFDM systems such as 802.11a/n or 802.16, there is a typical OFDM block structure. A OFDM block consists of a set of preamble symbols used for acquisition, synchronization and channel estimation, followed by a set of serially concatenated OFDM data symbols. User mobility gives rise to a considerable variation of the mobile channel during one OFDM block what causes *outdated channel information* in certain OFDM symbols if there is no appropriate channel tracking. To be precise, during the time period λ between channel estimation and OFDM symbol reception the channel changes in a way that the estimated channel information used for equalization does not fit the actual channel anymore. If there is no channel tracking at the receiver side, our aim is it to incorporate the effect of outdated channel information into the analytical BER derivation. Therefore we have to define the auto-correlation properties of channel coefficients H_l . The auto-correlation coefficient of subcarrier l is defined as follows

$$r_H(l, \lambda) = \frac{E\{H_l(t)H_l^*(t + \lambda)\}}{\sigma_H^2} \quad (11)$$

Applying Eq.(2) we get

$$E\{H_l(t)H_l^*(t + \lambda)\} = E\left\{\sum_{\tau=0}^{L-1}\sum_{\nu=0}^{L-1}h(\tau, t)h^*(\nu, t + \lambda) \times e^{-2\pi i \frac{(\tau-\nu)\lambda}{N}}\right\} \quad (12)$$

When assuming uncorrelated channel taps it follows

$$E\{H_l(t)H_l^*(t + \lambda)\} = \sum_{\tau=0}^{L-1} r_h(\tau, \lambda) \sigma_\tau^2 \quad (13)$$

For sake of simplicity, it is assumed that all channel taps have the same auto-correlation coefficient $r_h(\tau, \lambda) = r_h(\lambda) \quad \forall 0 \leq \tau \leq L - 1$. Together with the relation $\sum_{\tau=0}^{L-1} \sigma_\tau^2 = \sigma_H^2$, we obtain

$$r_H(l, \lambda) = r_h(\lambda) \quad (14)$$

For the numerical evaluation of the analytical BER results in Sec.V, the time selectivity of the complex Gaussian channel taps was modeled as follows

$$h(\tau, t + \lambda) = r_h(\tau, \lambda)h(\tau, t) + v_{\tau, \lambda}$$

with

$$E\{|h(\tau, t)|^2\} = E\{|h(\tau, t + \lambda)|^2\} = \sigma_\tau^2$$

where $v_{\tau, \lambda}$ is a complex Gaussian RV with variance $\sigma_{v_{\tau, \lambda}}^2 = \sigma_\tau^2(1 - |r_h(\tau, \lambda)|^2)$ and $E\{h(\tau, t)v_{\tau, \lambda}^*\} = 0$. For sake of simplicity, it is assumed that the channel is stationary during one OFDM symbol but changes from symbol to symbol in the above defined manner.

III. PROBABILITY DENSITY FUNCTIONS ANALYSIS

The author of [7] has suggested a correlation model regarding channel estimation for single-carrier systems and derived the correspondent symbol error rate and bit error rate of QAM-modulated signals transmitted in flat Rayleigh and Ricean channels. In this section, a short review of the contribution of [7] will be given in order to further extend these results to OFDM systems for time and frequency selective fading channels with both carrier frequency offset and channel estimation error. The single-carrier transmission model without carrier frequency offset for flat Rayleigh fading channels can be written as

$$y = hx + w \quad (15)$$

where y , h , x and w denote the complex baseband representation of the received signal, the channel coefficient, the transmitted data and the Gaussian noise with variance σ_w^2 respectively. In [7], the channel estimate $\hat{h}$ is assumed to be biased and used for zero forcing equalization as follows

$$z = \frac{y}{\hat{h}} \quad \text{with} \quad \hat{h} = \alpha h + \nu \quad (16)$$

where α denotes the multiplicative bias in the channel estimates and ν is a zero-mean complex Gaussian noise with variance σ_ν^2 . The channel coefficient h and Gaussian noise ν are assumed to be uncorrelated.

In [7], the joint PDF of the decision variable $z = z_r + jz_i$ in case of transmit symbol x is derived in cartesian coordinates and can be written as

$$f_{Z|X}(z|x) = \frac{a^2(x)}{\pi(|z - b(x)|^2 + a^2(x))^2} \Big|_{z=z_r+jz_i} \quad (17)$$

The PDF mainly depends on the complex parameter $b(x)$ given by [7]

$$\begin{aligned} b(x) &= \Re\{b\} + j\Im\{b\} = b_r(x) + jb_i(x) \\ &= x \left(\frac{\alpha^* r_h(\lambda) \sigma_h^2}{|\alpha|^2 \sigma_h^2 + \sigma_\nu^2} \right) \end{aligned} \quad (18)$$

and the real parameter $a(x)$ that can be written as [7]

$$\begin{aligned} a^2(x) &= |x|^2 \frac{|\alpha|^2 \sigma_h^4 (1 - r_h^2(\lambda)) + \sigma_\nu^2 \sigma_h^2}{(|\alpha|^2 \sigma_h^2 + \sigma_\nu^2)^2} \\ &\quad + \frac{\sigma_w^2}{|\alpha|^2 \sigma_h^2 + \sigma_\nu^2} \end{aligned} \quad (19)$$

Additionally, the primitive of Eq.(17) with $z = z_r + jz_i$ is given by [7]

$$\begin{aligned} F_{Z|X}(z|x) &= \frac{(z_i - b_i(x)) \arctan\left(\frac{z_r - b_r(x)}{\sqrt{a^2(x) + (z_i - b_i(x))^2}}\right)}{2\pi \sqrt{a^2(x) + (z_i - b_i(x))^2}} \\ &\quad + \frac{(z_r - b_r(x)) \arctan\left(\frac{z_i - b_i(x)}{\sqrt{a^2(x) + (z_r - b_r(x))^2}}\right)}{2\pi \sqrt{a^2(x) + (z_r - b_r(x))^2}} \end{aligned} \quad (20)$$

The result of (20) can be used to calculate the bit error rate of a given M-QAM constellation. In a M-QAM constellation there are $M \log(M)$ different possible bit positions with respect

to the M-QAM constellation. For each constellation point x_m , where m denotes the constellation point index, the parameters $b(x_m) \equiv b_{m,r} + jb_{m,i}$ and parameter $a^2(x_m) \equiv a_m^2$ has to be calculated separately for evaluating the bit error rate using Eq.(20) and Eq.(21). The probability of an erroneous bit with respect to the m -th QAM symbol x_m can be calculated by using the primitive of conditional PDF Eq.(20) and an appropriate decision region $B_{m,\nu}$ for the bit position used for double integration. Hence, the bit error probability $P_b(x_m)$ takes the form

$$P_b(x_m) = \frac{1}{ld(M)} \sum_{\nu=1}^{ld(M)} [[F_{Z|X}(z|x_m)]]_{B_{m,\nu}} \quad (21)$$

Finally, the bit error probability can be obtained by averaging over all possible constellation points, when assuming equal probable M-QAM symbols.

IV. OFDM BIT ERROR RATE ANALYSIS

In this section, the derivation of the bit error rates of OFDM systems with carrier frequency offset, I/Q imbalance and channel estimation error in Rayleigh frequency and time selective fading channels will be given. The central idea of our BER derivation is to map the OFDM system model of Sec.II to the PDF method given in Sec.III. To be precise, we have to map the OFDM system model to the parameters α , a^2 (Eq.(19)) and b^2 (Eq.(18)) as explained below.

A. Mathematical Derivation

Firstly, we can rewrite the channel estimates of subcarrier l in Eq.(6) with respect to the frequency selective fading characteristic given in Eq.(10) to be

$$\hat{H}_l = I(0)H_l \left(1 + \frac{\sum_{k \neq l} r_{k,l} X_{P,k} I(k-l)}{I(0)X_{P,l}} + e \frac{K_l \sum_m r_{m,l}^* X_{P,m}^* I^*(m+l)}{I(0)X_{P,l}} \right) + \tilde{v}_l \quad (22)$$

where e denotes the term $e^{-j2\phi_l}$. This comes due to the fact that the complex Gaussian Channel coefficient can be written as $H_l = |H_l|e^{j\phi_l}$. Hence we have $\frac{H_l^*}{H_l} = e^{-j2\phi_l} = e$, where ϕ_l is a equally distributed RV in the interval $[-\pi : \pi]$. From Eq.(22) we obtain an Eq.(16)-like expression as follows

$$\hat{H}_l = \tilde{\alpha}_l \tilde{H}_l + \tilde{v}_l \quad (23)$$

by defining *effective channel* $\tilde{H}_l = I(0)H_l$ and *effective bias* $\tilde{\alpha}_l$ as

$$\tilde{\alpha}_l = 1 + \frac{\sum_{k \neq l} r_{k,l} X_{P,k} I(k-l) + e K_l \sum_m r_{m,l}^* X_{P,m}^* I^*(m+l)}{I(0)X_{P,l}}$$

where $\tilde{\alpha}_l$ is a stochastic quantity with given subcarrier index l , a set of deterministic preamble symbols $X_{P,k}$, a fixed predetermined frequency offset, a given IRR constant K_l and RV $e = e^{j2\phi_l}$. Hence, in the following we write $\tilde{\alpha}_l(\phi_l)$ to emphasize the stochastic nature with respect to ϕ_l .

The AWGN part $\tilde{v}_l$ of the channel estimate can be written as

$$\begin{aligned} \tilde{v}_l = & W'_l + K_l W'_{-l} + \frac{\sum_{k \neq l} X_{P,k} V_{k,l} I(k-l)}{X_{P,l}} \\ & + K_l \frac{\sum_m X_{P,m}^* V_{m,l}^* I^*(m+l)}{X_{P,l}} \end{aligned} \quad (24)$$

For $\sigma_{\tilde{v}_l}^2$, which represents the AWGN variance of the channel estimates, we have

$$\begin{aligned} \sigma_{\tilde{v}_l}^2 = & \sum_{k \neq l} \sum_{n \neq l} X_{P,k} X_{P,n}^* I(k-l) I^*(n-l) \\ & \times (r_{k,n} - r_{k,l} r_{n,l}^* \sigma_H^2) \\ & + |K_l|^2 \sum_k \sum_n X_{P,k}^* X_{P,n} I^*(k+l) I(n+l) \\ & \times (r_{k,n}^* - r_{k,l}^* r_{n,l} \sigma_H^2) \\ & + \sigma_W^2 (1 + |K_l|^2) \end{aligned} \quad (25)$$

Applying the same method as above for Eq.(3) and Eq.(4), the same definition of *effective channel* $\tilde{H}_l$ can be used to obtain an Eq.(15)-like expression as follows

$$Y_l = \tilde{H}_l \left(X_l + \frac{\sum_{k \neq l} r_{k,l} X_k I(k-l)}{I(0)} \right) + \tilde{W}_l \quad (26)$$

$$= \tilde{H} \tilde{X}_l + \tilde{W}_l \quad (27)$$

Given Eq.(26) and Eq.(27), the *effective symbol* $\tilde{X}_l$ can be defined, that is no longer a deterministic value but a stochastic quantity due to i.i.d. data symbols on subcarriers $k \neq l$. Giving a certain transmit symbol X_l and assuming randomly transmitted data signals X_k with $k \neq l$, we can decompose the effective symbol $\tilde{X}_l$ as $\tilde{X}_l = X_l + J_l$, which shows the stochastic nature of $\tilde{X}_l$ due to the random interference part J_l due to ICI and I/Q imbalance. Applying the central limit theorem, we assume that the interference J_l term is a complex zero-mean Gaussian random variable $J_l = p + jq$. The mutual uncorrelated real and imaginary parts p and q have the same variance for all constellation points

$$\sigma_{J_l}^2 = \frac{\sum_{k \neq l} |I(k-l)|^2 |r_{k,l}|^2 + |K_l|^2 \sum_m |I(m+l)|^2 |r_{m,l}|^2}{2|I(0)|^2}$$

According to Eq.(18) and Eq.(19), we calculate the parameters $b_l = b_{l,r} + jb_{l,i}$ and a_l^2 for M-QAM effective data symbols $\tilde{X}_l$ on subcarrier l in frequency and time selective fading channels.

$$\begin{aligned} b_l(\tilde{X}_l, \phi_l) = & \tilde{X}_l \left(\frac{\tilde{\alpha}_l^*(\phi_l) r_h(\lambda) \sigma_{\tilde{H}}^2}{|\tilde{\alpha}_l(\phi_l)|^2 \sigma_{\tilde{H}}^2 + \sigma_{\tilde{v}}^2} \right) \\ a_l^2(\tilde{X}_l, \phi_l) = & |\tilde{X}_l|^2 \frac{|\tilde{\alpha}_l(\phi_l)|^2 \sigma_{\tilde{H}}^4 (1 - r_h^2(\lambda)) + \sigma_{\tilde{v}_l}^2 \sigma_{\tilde{H}_l}^2}{(\sigma_{\tilde{H}_l}^2)^2} + \frac{\sigma_{\tilde{W}_l}^2}{\sigma_{\tilde{H}_l}^2} \end{aligned} \quad (28)$$

where $\sigma_{\tilde{H}_l}^2 = |I(0)|^2 \sigma_H^2$ and $\sigma_{\tilde{H}_l}^2 = |\tilde{\alpha}_l(\phi_l)|^2 |I(0)|^2 \sigma_H^2 + \sigma_{\tilde{v}_l}^2$. From Eqs.(28) one can observe, that the parameter $\sigma_{\tilde{W}_l}^2$ have to be calculated exactly to obtain reliable results. The terms $\tilde{W}_l$ represents the *effective noise* of the received signal that consists of AWGN parts W_l , W_{-l} and ICI parts respectively. If we substitute Eq.(3) and Eq.(10) into Eq.(4), we get

$$\begin{aligned}\tilde{W}_l &= W_l + K_l W_{-l} + \sum_{k \neq l} X_k V_{k,l} I(k-l) \\ &\quad + K_l \sum_m X_m^* V_{m,l}^* I^*(m+l)\end{aligned}\quad (29)$$

For an exact expression of $\sigma_{\tilde{W}_l}^2$ we take Eq.(29), $\sigma_{V_{k,l}}^2 = \sigma_H^2(1 - |r_{k,l}|^2)$ together with the assumptions of mutually uncorrelated data symbols and obtain

$$\begin{aligned}\sigma_{\tilde{W}_l}^2 &= \sigma_W^2(1 + |K_l|^2) + \sigma_H^2 \sum_{k \neq l} |I(k-l)|^2(1 - |r_{k,l}|^2) \\ &\quad + |K_l|^2 \sigma_H^2 \sum_{m \neq l} |I(m+l)|^2(1 - |r_{m,l}|^2)\end{aligned}\quad (30)$$

Subsequently, the bit error rate on subcarrier l for the m -th constellation point can be expressed using Eq.(20) and Eq.(21) by the following triple integral

$$\begin{aligned}\bar{P}_b(X_{m,l}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{P_b(X_{m,l} + p + jq, \phi)}{2\pi \tilde{\sigma}_{J_l}^2} \\ &\quad \times e^{-\frac{p^2 + q^2}{2\tilde{\sigma}_{J_l}^2}} dp dq d\phi\end{aligned}\quad (31)$$

Finally, to obtain the general bit error rate we have to average Eq.(31) over all data subcarriers and M-QAM constellation

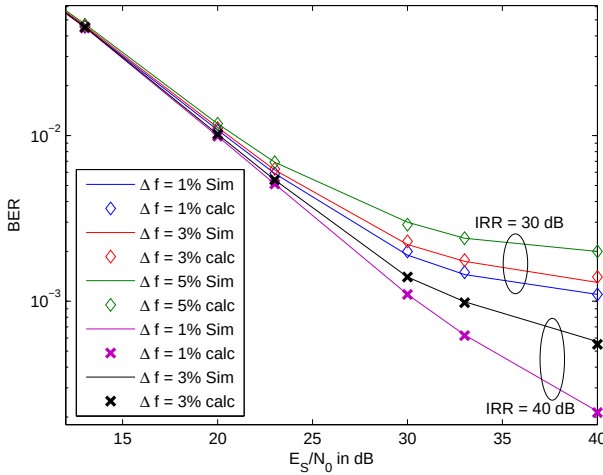


Fig. 1. Comparison of simulated and analytical uncoded BER vs. SNR for 4-QAM OFDM under residual CFO in static channel environment

V. NUMERICAL RESULTS

In numerical examples, we consider a OFDM system as described in Sec.II. Fig.1, present the calculated and simulated bit error rates vs. SNR (E_s/N_0) with given carrier frequency offset Δf (in % subcarrier spacing) for static non time variant 8 taps exponential PDP frequency selective channels with $D = 7$ (see Sec.II). The results illustrate that our analysis can approximate the simulative performance very accurately, if the power delay profile of fading channels, the image rejection ratio of the direct conversion receiver and carrier frequency offset are known. In Fig.2 we also use a fixed Δf of 3% to investigate 4-QAM BER for time variant exponential 8 taps PDP channels characterized by the channel tap auto correlation coefficients $r_h(\lambda)$.

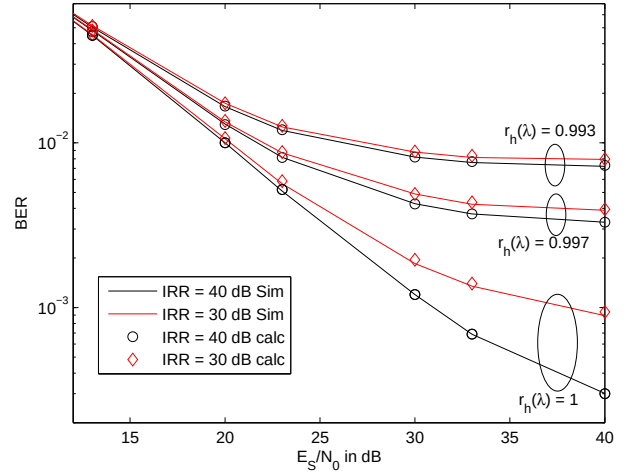


Fig. 2. Comparison of simulated and analytical uncoded BER vs. SNR for 4-QAM OFDM with residual CFO of 3% under time selective channel conditions

VI. CONCLUSIONS

In this paper we show how to analytically evaluate the uncoded bit error probability/rate of OFDM systems subject to carrier frequency offset, channel estimation error, outdated channel information and flat receiver I/Q imbalance in Rayleigh frequency and time selective mobile fading channels. We show that the bit error probability can be calculated exactly when using a priori knowledge of mobile channel power delay profile, image rejection ratio and receiver CFO. Therefore, our analysis provides a useful and necessary tool for system engineers to cross-reference the performance of OFDM synchronization/acquisition algorithms with appropriate uncoded bit error rates.

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